THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A chord is at distance 7 cm from the centre of a circle of radius 25 cm. Its length (in cm) is: 1

(a) 50 **(b)** 47 **(c)** 48 **(d)** 49

2. The area of a triangle with sides 13 cm, 14 cm and 15 cm is (cm^2): 1

(a) 83 **(b)** 84 **(c)** 85 **(d)** 86

3. A die is thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. The experimental probability of getting 2 is: 1

(a) $\frac{11}{51}$ **(b)** $\frac{6}{25}$ **(c)** $\frac{11}{50}$ **(d)** $\frac{39}{50}$

4. If $t_n = 2n^2 + 3$, then t_5 is: 1

(a) 55 **(b)** 52 **(c)** 54 **(d)** 53

5. The radius of a circle is 25 cm and a chord is 40 cm long. The distance of the chord from the centre (in cm) is: 1

(a) 17 **(b)** 14 **(c)** 16 **(d)** 15

6. The semi-perimeter of a triangle with sides 13 cm, 14 cm and 15 cm is (cm): 1

(a) 23 **(b)** 22 **(c)** 20 **(d)** 21

7. In a survey of 500 families, 140 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{8}{25}$ **(b)** $\frac{18}{25}$ **(c)** $\frac{7}{25}$ **(d)** $\frac{7}{26}$

8. Which term of the sequence $t_n = 3n + 2$ is 53? 1

(a) 19th term **(b)** 17th term **(c)** 18th term **(d)** 16th term

9. The chord AB subtends 74° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 27° **(b)** 47° **(c)** 57° **(d)** 37°

10. The area of a circle with circumference 88 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 616 **(b)** 615 **(c)** 617 **(d)** 618

11. A letter is chosen at random from the word EDUCATION. The probability that it is a vowel is: 1

(a) $\frac{4}{9}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{5}{9}$

12. A sequence has $t_1 = 4$ and $t_{n+1} = t_n + 3$. Then t_6 equals: 1

(a) 19 **(b)** 21 **(c)** 20 **(d)** 18

13. The chord AB subtends 66° at the centre. The angle it subtends at a point on the minor arc is: 1

(a) 167° **(b)** 137° **(c)** 147° **(d)** 157°

14. A wheel of radius 28 cm (take $\pi = \frac{22}{7}$) makes one full turn. The distance covered (in cm) is: 1

(a) 175 **(b)** 176 **(c)** 177 **(d)** 178

15. Three coins are tossed together. The probability of getting exactly one head is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{5}{8}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. In a circle of radius 26 cm, a chord is 10 cm from the centre. Find its length. 2

17. Find the area of a triangle with sides 13 cm, 14 cm, 15 cm. 2

18. A die was thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. Find the experimental probability of getting an even number. 2

19. A sequence is defined by $t_n = 3n + 1$. Find t_8 and decide whether 37 is a term. 2

20. A chord AB of a circle with centre O subtends an angle of 60° at O. Find the angle it subtends at a point C on the minor arc. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In triangle OAB with O the centre of a circle, $\angle AOB = 70^\circ$. Find $\angle OAB$ and the angle subtended by AB at a point of the major arc. 3

22. Find the area of the triangle with sides 13 cm, 14 cm and 15 cm, and the length of the altitude on the side of 15 cm. 3

23. A bag contains 4 red, 3 black and 5 white balls. A ball is drawn at random. Find the probability that it is 3

(i) red,

(ii) not red,

(iii) black or white.

24. Find which term of the AP $7, 13, 19, 25, \ldots$ equals 151. 3

OR

In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 10)^\circ$ and $\angle C = (3x - 5)^\circ$. Find x and $\angle A$. 3

25. Find the area of the quadrilateral with sides 4, 5, 7, 10 cm that is inscribed in a circle. 3

OR

In a survey of 200 students, 80 take the bus, 50 cycle and the rest walk. Find the probability that a randomly chosen student 3

(i) walks,

(ii) does not take the bus.

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SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre of a circle of radius 13 cm. Find the distance between them. **5**

OR

A circular track has inner radius 14 m and outer radius 21 m (take $\pi = \frac{22}{7}$). Find the area of the track and the cost of levelling it at ₹5 per m^2. **5**

27. A die was thrown 100 times; the number of times $1, \ldots, 6$ appeared was 15, 22, 17, 16, 14, 16. Find the experimental probability of **5**

(i) a prime number,

(ii) a number greater than 4,

(iii) a number less than 3, and check that all six relative frequencies add to 1.

OR

A taxi charges a fixed ₹200 and ₹40 per km. **5**

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 12 km.

END OF QUESTION PAPER — Total: 50 Marks

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